

Exact Second-Order Light Shifts of Alkali-Metal Ground States

Scalar, vector, and tensor components for a quantization axis along z

Hyperfine-resolved derivation; arbitrary polarization; Doppler averaging; connection to Mathur–Tang–Happer

Revision v1.4 – uses F' consistently for excited hyperfine levels

Main result. For a ground hyperfine Zeeman state $|F, m\rangle = |nS_{1/2}; F, m\rangle$, with z the quantization axis and with Zeeman mixing negligible, the Doppler-averaged dispersive ac Stark shift can be written

$$\overline{\Delta E_{Fm}} = -\frac{I_L}{2\epsilon_0 c} \left[\overline{\alpha}_F^s + C_z \overline{\alpha}_F^v \frac{m}{2F} - D_z \overline{\alpha}_F^T \frac{3m^2 - F(F+1)}{2F(2F-1)} \right]. \quad (1)$$

Here

$$C_z \equiv -i(\boldsymbol{\epsilon}^* \times \boldsymbol{\epsilon})_z = |\epsilon_{-1}|^2 - |\epsilon_{+1}|^2, \quad D_z \equiv 1 - 3|\epsilon_z|^2. \quad (2)$$

The overbar emphasizes that the hyperfine-resolved optical denominators have been thermally averaged over the velocity distribution. For a stationary atom, simply remove the overbars. For $F < 1$, the tensor term is absent.

1. Scope, assumptions, and definitions

(P1) “Exact” means exact within second-order electric-dipole perturbation theory for the specified set of atomic eigenstates. The counter-rotating term may be retained, the intermediate-state hyperfine energies are retained, and Doppler averaging is performed before summing over the distinct excited hyperfine components. This does not include higher-order light shifts, direct magnetic-dipole/electric-quadrupole optical couplings, or nonperturbative near-resonant dynamics.

(P2) The most rigorous form uses exact hyperfine eigenenergies and exact hyperfine reduced dipole matrix elements. The alkali-specific $6j$ reduction used later assumes that hyperfine coupling changes the level energies much more strongly than it changes the electronic radial wave functions.

(P3) Equation (1) is a diagonal expectation value. If the light-shift Hamiltonian has off-diagonal elements comparable to the Zeeman splittings, m is not a good eigenlabel and the full operator must be diagonalized.

1.1. Definitions

Symbol	Definition
$J = 1/2$	Electronic angular momentum of the alkali $nS_{1/2}$ ground state.
I	Nuclear spin.
F	Total angular momentum, $\mathbf{F} = \mathbf{I} + \mathbf{J}$.
m	Eigenvalue label of $F_z/\hbar$, with $m = -F, \dots, F$.
ω	Laboratory laser angular frequency.
ϵ	Complex unit polarization vector, $\epsilon^* \cdot \epsilon = 1$.
$\mathcal{E}$	Complex peak electric-field amplitude used below.
I_L	Cycle-averaged intensity, $I_L = \frac{1}{2}\epsilon_0 c \mathcal{E} ^2$.
$\omega_{F'F}$	Positive transition angular frequency from ground hyperfine level F to excited hyperfine level F' (with $n'J'$ suppressed when a single excited manifold is being discussed).
$\gamma_{F'F}$	Homogeneous optical-coherence linewidth of the $F \rightarrow F'$ optical transition.
u	One-dimensional most-probable thermal speed, $u = \sqrt{2k_B T/M}$.
$\overline{\alpha}_F^{s,v,T}$	Doppler-averaged conventional scalar, vector, and tensor polarizabilities.

2. Starting point: electric-dipole interaction and second-order shift

Take the real field to be

$$\mathbf{E}(t) = \frac{1}{2}\mathcal{E}\epsilon e^{-i\omega t} + \frac{1}{2}\mathcal{E}^*\epsilon^* e^{+i\omega t}, \quad I_L = \frac{1}{2}\epsilon_0 c |\mathcal{E}|^2, \quad (3)$$

with electric-dipole coupling

$$V(t) = -\mathbf{d} \cdot \mathbf{E}(t). \quad (4)$$

For the hyperfine-resolved derivation in this section, we consider one specified excited electronic/fine-structure manifold and suppress its $n'J'$ label. Thus $|F', m'\rangle$ always denotes an *excited hyperfine state*, whereas $|F, m\rangle$ denotes a ground hyperfine state. When several excited manifolds are included, the $n'J'$ labels are restored explicitly, as in Eqs. (8)–(10).

For a stationary ground state $|F, m\rangle$, second-order Floquet perturbation theory gives the dispersive shift

$$\Delta E_{Fm} = -\frac{|\mathcal{E}|^2}{4\hbar} \sum_{F', m'} \left[|\langle F'm' | \epsilon \cdot \mathbf{d} | Fm \rangle|^2 \operatorname{Re} \frac{1}{\omega_{F'F} - \omega - i\gamma_{F'F}/2} + |\langle Fm | \epsilon \cdot \mathbf{d} | F'm' \rangle|^2 \operatorname{Re} \frac{1}{\omega_{F'F} + \omega + i\gamma_{F'F}/2} \right]. \quad (5)$$

The first term is rotating; the second is counter-rotating. Here F' is the excited hyperfine angular momentum, and $\omega_{F'F}$ is the transition frequency from the ground hyperfine level F to the excited hyperfine level F' within the specified excited manifold.

The missing operator step between Eqs. (5) and (6a)

Equation (5) is only a *diagonal matrix element*. It gives the shift of one specified state $|F, m\rangle$, but it no longer contains the off-diagonal information needed to reconstruct an operator in the complete F manifold. Therefore Eq. (6a) does **not** follow directly from Eq. (5). One must retain the second-order *effective operator* before taking its expectation value.

Let

$$P_F = \sum_{m=-F}^F |Fm\rangle\langle Fm|, \quad P_{F'} = \sum_{m'=-F'}^{F'} |F'm'\rangle\langle F'm'|, \quad (5a)$$

and define the real (dispersive) rotating and counter-rotating resolvents

$$\begin{aligned} R_+(\omega) &= \sum_{F'} \text{Re} \left[\frac{1}{\omega_{F'F} - \omega - i\gamma_{F'F}/2} \right] P_{F'}, \\ R_-(\omega) &= \sum_{F'} \text{Re} \left[\frac{1}{\omega_{F'F} + \omega + i\gamma_{F'F}/2} \right] P_{F'}. \end{aligned} \quad (5b)$$

Then the dispersive second-order operator acting inside the ground F manifold is

$$H_{\text{LS}}^{(2,F)} = -\frac{|\mathcal{E}|^2}{4\hbar} P_F [(\boldsymbol{\epsilon}^* \cdot \mathbf{d}) R_+(\omega) (\boldsymbol{\epsilon} \cdot \mathbf{d}) + (\boldsymbol{\epsilon} \cdot \mathbf{d}) R_-(\omega) (\boldsymbol{\epsilon}^* \cdot \mathbf{d})] P_F. \quad (5c)$$

Taking $\langle Fm | \cdots | Fm \rangle$ in Eq. (5c) reproduces Eq. (5). Thus Eq. (5c), not Eq. (5), is the correct parent of Eq. (6a).

To expose the rotational structure, expand each dipole matrix element with the Wigner–Eckart theorem,

$$\langle F'm' | d_p | Fm \rangle = (-1)^{F'-m'} \begin{pmatrix} F' & 1 & F \\ -m' & p & m \end{pmatrix} \langle F' || d || F \rangle. \quad (5d)$$

For example, the rotating part of an arbitrary matrix element $\langle Fm_2 | H_{\text{LS}} | Fm_1 \rangle$ contains

$$\sum_{m'} \langle Fm_2 | d_{p'} | F'm' \rangle \langle F'm' | d_p | Fm_1 \rangle. \quad (5e)$$

The product in Eq. (5e) is a product of two rank-1 tensors. Racah recoupling of the two $3j$ symbols in Eq. (5d) gives

$$\sum_{m'} (\cdots) = \sum_{K=0}^2 \sum_{Q=-K}^K B_{FF'}^{(K)} \langle 1p', 1p | KQ \rangle \langle Fm_2 | T_Q^{(K)}(F) | Fm_1 \rangle. \quad (5f)$$

Here $B_{FF'}^{(K)}$ contains the reduced dipole matrix element and a Wigner $6j$ symbol. Its precise numerical normalization depends on the chosen normalization of $T_Q^{(K)}$; after the denominators are included it becomes the coefficient $A_F^{(K)}$ (or, equivalently, the reduced polarizability $\alpha_F^{(K)}$ of Eq. (8)). The important point is structural: because $1 \otimes 1 = 0 \oplus 1 \oplus 2$, only $K = 0, 1, 2$ can occur.

The polarization amplitudes carry the *same* Clebsch–Gordan coupling,

$$\{\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon}\}_{KQ} \equiv \sum_{p'p} \langle 1p', 1p | KQ \rangle (\boldsymbol{\epsilon}^*)_{p'} \epsilon_p, \quad (5g)$$

with the usual spherical-vector conjugation convention understood. Consequently the sums over p, p' in Eq. (5c) pair the rank- K light tensor with the rank- K atomic tensor. Rotational invariance requires their scalar product,

$$(\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon})^{(K)} \cdot T^{(K)}(F) = \sum_{Q=-K}^K (-1)^Q \{\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon}\}_{KQ} T_{-Q}^{(K)}(F). \quad (5h)$$

Collecting the spectral denominators, reduced dipole matrix elements, and the $6j$ factor into $A_F^{(K)}$ gives Eq. (6a). In other words, the logical chain is

$$\boxed{V = -\mathbf{d} \cdot \mathbf{E}} \longrightarrow \boxed{\text{second-order operator, Eq. (5c)}} \longrightarrow \boxed{\text{Wigner–Eckart + Racah recoupling, Eqs. (5d)–(5g)}} \longrightarrow \boxed{\text{irreducible scalar contraction, Eq. (6a)}}.$$

Do not confuse two unrelated factors of $1/2$. The overall $-\frac{1}{4}|\mathcal{E}|^2\alpha$ form contains the familiar factor associated with an induced dipole together with optical-cycle averaging for the peak-field convention in Eq. (3). The later factors $m/(2F)$ and $[3m^2 - F(F+1)]/[2F(2F-1)]$ are instead part of the *conventional angular normalization*; they can be moved into a redefinition of α_F^v or α_F^T without changing any observable.

3. Irreducible decomposition: ranks $K = 0, 1, 2$

Because the interaction is second order in a rank-1 dipole operator,

$$1 \otimes 1 = 0 \oplus 1 \oplus 2. \quad (6)$$

Thus the effective light-shift operator separates into scalar ($K = 0$), vector ($K = 1$), and tensor ($K = 2$) pieces.

3.1. General irreducible operator form – the origin of the conventional Hamiltonian

Within a fixed hyperfine manifold F , the most compact rotationally invariant form is a scalar contraction of a rank- K light tensor with a rank- K atomic tensor:

$$\begin{aligned} H_{\text{LS}}^{(F)} &= -\frac{I_L}{2\epsilon_0 c} \sum_{K=0}^2 A_F^{(K)} [(\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon})^{(K)} \cdot T^{(K)}(F)] \\ &= -\frac{I_L}{2\epsilon_0 c} \sum_{K=0}^2 \sum_{q=-K}^K A_F^{(K)} (-1)^q \{\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon}\}_{Kq} T_{-q}^{(K)}(F). \end{aligned} \quad (6a)$$

This is the explicit scalar-product version of the shorthand used in Chat 2, Eq. (1), $H_{\text{LS}} = \sum_{Kq} A_F^{(K)} \{\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon}\}_{Kq} T_q^{(K)}(F)$. The apparent difference in $q \leftrightarrow -q$ and the factor $(-1)^q$ is only the standard definition of a spherical-tensor scalar product; these factors can instead be included in the definition of the atomic tensor components.

Equation (6a) is the conceptual parent of the conventional Cartesian Hamiltonian in Eq. (17). The three ranks have the following Cartesian content:

$$K = 0 : \quad (\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon})^{(0)} T^{(0)} \longrightarrow \text{identity in the fixed-}F \text{ manifold}, \quad (6b)$$

$$K = 1 : \quad (\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon})^{(1)} \cdot T^{(1)} \longrightarrow -i(\boldsymbol{\epsilon}^* \times \boldsymbol{\epsilon}) \cdot \mathbf{F}, \quad (6c)$$

$$K = 2 : \quad (\boldsymbol{\epsilon}^* \otimes \boldsymbol{\epsilon})^{(2)} \cdot T^{(2)} \longrightarrow 3[(\boldsymbol{\epsilon}^* \cdot \mathbf{F})(\boldsymbol{\epsilon} \cdot \mathbf{F}) + (\boldsymbol{\epsilon} \cdot \mathbf{F})(\boldsymbol{\epsilon}^* \cdot \mathbf{F})] - 2\mathbf{F}^2. \quad (6d)$$

The arrows in Eqs. (6b)–(6d) mean “same rotational tensor structure, after choosing a normalization.” They are not equalities before the Clebsch–Gordan normalization factors are included. Those normalization factors, together with the reduced atomic response $A_F^{(K)}$, are precisely what are redistributed into the conventional polarizabilities α_F^s , α_F^v , and α_F^T below.

A useful Cartesian way to see the rank-2 structure is to define the symmetric-traceless spin-alignment tensor

$$Q_{ij}^{(F)} \equiv \frac{3(F_i F_j + F_j F_i) - 2\delta_{ij} \mathbf{F}^2}{2F(2F-1)}. \quad (6e)$$

Then the tensor term of Eq. (17) is simply proportional to $\epsilon_i^* \epsilon_j Q_{ij}^{(F)}$. Thus Eq. (17) is not a new ansatz: it is Eq. (6a) rewritten from the orthonormal spherical-tensor basis into familiar Cartesian orientation ($K = 1$) and alignment ($K = 2$) operators.

Within a fixed hyperfine manifold F , define reduced rank- K polarizabilities $\alpha_F^{(K)}$. A commonly used conversion to the conventional scalar/vector/tensor coefficients is

$$\alpha_F^s = \frac{1}{\sqrt{3(2F+1)}} \alpha_F^{(0)}, \quad (7a)$$

$$\alpha_F^v = -\sqrt{\frac{2F}{(F+1)(2F+1)}} \alpha_F^{(1)}, \quad (7b)$$

$$\alpha_F^T = -\sqrt{\frac{2F(2F-1)}{3(F+1)(2F+1)(2F+3)}} \alpha_F^{(2)}. \quad (7c)$$

Equation (7c) exists only for $F \geq 1$.

3.2. Hyperfine-resolved rank- K sum

Restoring the excited electronic/fine-structure labels, write an excited alkali hyperfine state as $|n'P_{J'}F', m'\rangle$. The hyperfine angular momentum is still denoted simply by F' . The reduced rank- K response may then be written in the standard Wigner-6j form

$$\alpha_F^{(K)}(\omega) = (-1)^{K+F+1} \sqrt{2K+1} \sum_{n'J'F'} (-1)^{F'} \left\{ \begin{matrix} 1 & K & 1 \\ F & F' & F \end{matrix} \right\} \frac{|\langle n'P_{J'}F' \| d \| nS_{1/2}F \rangle|^2}{\hbar} D_K(\omega_{n'J'F',F}, \omega, \gamma_{n'J'F',F}), \quad (8)$$

where

$$D_K(\omega_0, \omega, \gamma) = \text{Re} \left[\frac{1}{\omega_0 - \omega - i\gamma/2} + \frac{(-1)^K}{\omega_0 + \omega + i\gamma/2} \right]. \quad (9)$$

The precise phase convention of the reduced matrix elements and 6j symbol must be used consistently; observable shifts are convention independent.

3.3. Alkali-specific hyperfine reduction

If the hyperfine interaction does not appreciably modify the electronic radial wave functions,

$$|\langle n'P_{J'}F' \| d \| nS_{1/2}F \rangle|^2 = (2F'+1)(2F+1) \left\{ \begin{matrix} F & 1 & F' \\ J' & I & 1/2 \end{matrix} \right\}^2 |\langle n'P_{J'} \| d \| nS_{1/2} \rangle|^2. \quad (10)$$

Substitution of Eq. (10) into Eq. (8) gives the practical hyperfine-resolved alkali sum. The essential point for the tensor shift is that each F' retains its own transition frequency and hence its own dispersive denominator.

4. Why C, D differ from the raw irreducible polarization tensors

The light polarization itself can be decomposed into irreducible tensors. With the spherical-basis convention

$$\epsilon_{-1} = \frac{\epsilon_x - i\epsilon_y}{\sqrt{2}}, \quad \epsilon_0 = \epsilon_z, \quad \epsilon_{+1} = -\frac{\epsilon_x + i\epsilon_y}{\sqrt{2}}, \quad (11)$$

the $q = 0$ rank-1 and rank-2 components are

$$\{\epsilon^* \otimes \epsilon\}_{10} = \frac{|\epsilon_{+1}|^2 - |\epsilon_{-1}|^2}{\sqrt{2}}, \quad (12a)$$

$$\{\epsilon^* \otimes \epsilon\}_{20} = \frac{3|\epsilon_0|^2 - 1}{\sqrt{6}}. \quad (12b)$$

The conventional Cartesian polarization parameters used in Eq. (1) are merely rescaled versions:

$$\boxed{C_z = -\sqrt{2} \{\epsilon^* \otimes \epsilon\}_{10}}, \quad \boxed{D_z = -\sqrt{6} \{\epsilon^* \otimes \epsilon\}_{20}}. \quad (13)$$

Thus C_z, D_z contain the same physical polarization information as the irreducible tensors; the square-root Clebsch–Gordan normalization has simply been absorbed into the conventional definitions of the polarizabilities/operator factors.

4.1. Why the visible vector 1/2 is conventional

The normalized atomic orientation is $m/F \in [-1, 1]$. The conventional vector term is written

$$\Delta E_{Fm}^{(v)} = -\frac{I_L}{2\epsilon_0 c} C_z \alpha_F^v \frac{m}{2F}. \quad (14)$$

For $|C_z| = 1$, the two stretched states receive opposite shifts $\pm \alpha_F^v/2$ times the common intensity prefactor, so their full vector splitting is proportional to α_F^v . One may instead define $\tilde{\alpha}_F^v = \alpha_F^v/2$ and write $\tilde{\alpha}_F^v(m/F)$; the physics is identical. Therefore this 1/2 is not the induced-dipole factor.

4.2. The tensor normalization and P_2

Using $D_z = 1 - 3|\epsilon_z|^2$, the tensor angular factor can be rearranged as

$$-D_z \frac{3m^2 - F(F+1)}{2F(2F-1)} = \underbrace{\frac{3|\epsilon_z|^2 - 1}{2}}_{P_2 \text{ of the light axis}} \underbrace{\frac{3m^2 - F(F+1)}{F(2F-1)}}_{\text{normalized atomic alignment}}. \quad (15)$$

For linear polarization at an angle θ to z , the first factor becomes

$$P_2(\cos \theta) = \frac{3 \cos^2 \theta - 1}{2}, \quad (16)$$

and for a stretched state $m = \pm F$ the second factor equals 1. This is the cleanest interpretation of the conventional tensor normalization.

5. Fixed- F light-shift operator and the z -axis result

Starting from the irreducible scalar contraction in Eq. (6a), insert the conventional coefficient definitions in Eq. (7), use the rank-1 and rank-2 Cartesian structures in Eqs. (6c)–(6e), and absorb the remaining Clebsch–Gordan normalization constants into $\alpha_F^{s,v,T}$. The dispersive operator in a fixed F manifold then becomes

$$H_{\text{LS}}^{(F)} = -\frac{I_L}{2\epsilon_0 c} \left[\alpha_F^s - i\alpha_F^v \frac{(\epsilon^* \times \epsilon) \cdot \mathbf{F}}{2F} + \alpha_F^T \frac{3[(\epsilon^* \cdot \mathbf{F})(\epsilon \cdot \mathbf{F}) + (\epsilon \cdot \mathbf{F})(\epsilon^* \cdot \mathbf{F})] - 2\mathbf{F}^2}{2F(2F-1)} \right]. \quad (17)$$

Taking $\langle F, m | \cdots | F, m \rangle$ gives Eq. (1). This explicitly closes the chain

$$\begin{aligned} \text{second-order dipole interaction} &\longrightarrow \text{irreducible form, Eq. (6a)} \\ &\Downarrow \\ \text{conventional Cartesian operator, Eq. (17)} &\longrightarrow \text{z-axis level shift, Eq. (1).} \end{aligned}$$

With the present spherical convention, $C_z = +1$ for the field component that drives $\Delta m = +1$ absorption.

5.1. Useful polarization cases

Polarization	$ \epsilon_z ^2$	C_z	D_z
Pure π (linear along z)	1	0	-2
Linear transverse (e.g. x)	0	0	+1
Pure σ^+ in xy plane	0	+1	+1
Pure σ^- in xy plane	0	-1	+1

Circular polarization can therefore carry rank-0, rank-1, and rank-2 polarization components simultaneously.

6. Doppler effect: average the denominators before the hyperfine sum

For an atom with velocity component v_z along the light wave vector, the optical frequency in the atom's frame is, to first order in v/c ,

$$\omega \longrightarrow \omega - kv_z. \quad (18)$$

The one-dimensional Maxwell distribution is

$$f_T(v_z) = \frac{1}{\sqrt{\pi}u} \exp(-v_z^2/u^2), \quad u = \sqrt{\frac{2k_B T}{M}}. \quad (19)$$

For each optical transition, replace Eq. (9) by the Doppler-averaged response

$$\boxed{\bar{D}_K(\omega_0, \omega, \gamma; T) = \int_{-\infty}^{\infty} f_T(v_z) \operatorname{Re} \left[\frac{1}{\omega_0 - (\omega - kv_z) - i\gamma/2} + \frac{(-1)^K}{\omega_0 + (\omega - kv_z) + i\gamma/2} \right] dv_z.} \quad (20)$$

Then the Doppler-averaged $\bar{\alpha}_F^{(K)}$ is obtained from Eq. (8) simply by the replacement $D_K \rightarrow \bar{D}_K$. This is the natural way to include Doppler broadening while preserving the scalar/vector/tensor decomposition.

6.1. Plasma-dispersion (Faddeeva) form of the rotating term

Define

$$x_{F'F} = \frac{\omega - \omega_{F'F}}{ku}, \quad y_{F'F} = \frac{\gamma_{F'F}}{2ku}, \quad (21)$$

and the Fried–Conte plasma-dispersion function

$$Z(\zeta) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{t - \zeta} dt, \quad \Im \zeta > 0. \quad (22)$$

The rotating contribution becomes

$$\int f_T(v_z) \operatorname{Re} \frac{dv_z}{\omega_{F'F} - \omega + kv_z - i\gamma_{F'F}/2} = \frac{1}{ku} \operatorname{Re} Z(x_{F'F} + iy_{F'F}). \quad (23)$$

This is the modern form of the Doppler-broadened dispersive profile used by Mathur, Tang, and Happer. Their dimensionless x and y are written in ordinary-frequency, molar-mass, and Gaussian-unit conventions, but encode the same velocity average.

Critical point for the tensor shift. The average in Eq. (20) must be evaluated separately for every excited hyperfine state F' , because each has a different $\omega_{F'F}$. If the excited hyperfine splittings are much smaller than the Doppler width, the functions $Z(x_{F'F} + iy)$ become nearly equal. The rank-2 angular coefficients then cancel for an alkali $S_{1/2}$ ground state. If the F' components are spectrally resolved enough to receive different dispersive weights, the cancellation is incomplete and a hyperfine-mediated tensor shift survives.

7. Connection to Mathur–Tang–Happer (1968)

Mathur, Tang, and Happer write the effective ground-state light-shift operator schematically as

$$\delta E = \delta E_0 + \hbar \delta A \mathbf{I} \cdot \mathbf{J} - \boldsymbol{\mu} \cdot \delta \mathbf{H} + \delta E_2, \quad (24)$$

where δE_2 is explicitly identified as the tensor light-shift operator. Therefore, for one ground-state Zeeman sublevel,

$$\boxed{\delta \nu_T(F, m) = \frac{\langle F, m | \delta E_2 | F, m \rangle}{h}}. \quad (25)$$

This is the same physical object as the tensor term in Eq. (1), expressed in their tensor-operator normalization.

For the specific ^{87}Rb field-independent “0–0” hyperfine transition, their Eq. (20) is instead the *difference* of two tensor level shifts,

$$\delta \nu_t = \frac{\langle 2, 0 | \delta E_2 | 2, 0 \rangle - \langle 1, 0 | \delta E_2 | 1, 0 \rangle}{h}. \quad (26)$$

Thus Eq. (25) is a single-level tensor shift, whereas Eq. (26) is a transition-frequency shift.

7.1. Their rank-2 polarization factor versus D_z

Mathur et al. define a rank-2 polarization component (their notation E_0^2 ; the superscript denotes tensor rank, not a squared field amplitude)

$$E_0^2 = \frac{1}{\sqrt{6}} - \sqrt{\frac{3}{2}} |e_z|^2 = \frac{1 - 3|e_z|^2}{\sqrt{6}}. \quad (27)$$

Therefore

$$\boxed{D_z = \sqrt{6} E_0^2}. \quad (28)$$

This is the same normalization issue as Eq. (13): Mathur keeps the spherical-tensor $1/\sqrt{6}$ normalization, while the conventional D_z removes it.

7.2. Broad-spectrum light

For a lamp or other light source with spectral energy-flux density $\Phi(\omega)$, Mathur’s response-function structure is naturally written

$$\delta \nu_T = \mathcal{P}^{(2)} \int S_2(\omega) \Phi(\omega) d\omega, \quad (29)$$

where $\mathcal{P}^{(2)}$ is the rank-2 polarization factor and S_2 contains the hyperfine angular coefficients multiplied by $\text{Re } Z(x + iy)$. For a monochromatic laser, $\Phi(\omega)$ is proportional to a delta function and Eq. (29) reduces to evaluating the Doppler-broadened polarizability at the laser frequency.

8. Why an alkali $S_{1/2}$ ground state can have a tensor shift

At the purely electronic fine-structure level, $J = 1/2$ cannot support a rank-2 electronic polarizability. If the excited hyperfine splittings are neglected in the optical denominators, the sum over F' collapses and the tensor contribution cancels:

$$\alpha_F^T = 0 \quad (\text{unresolved excited hyperfine structure, ideal } S_{1/2} \text{ limit}). \quad (30)$$

With hyperfine-resolved denominators, however, each excited F' receives a different spectral weight. Doppler broadening determines how different those weights remain after thermal averaging. This is why Mathur et al. emphasize that the tensor shift is small when excited-state hyperfine splittings are small compared with the Doppler width, but is appreciable in Rb and Cs when the first excited hyperfine structure is sufficiently resolved.

9. Near-resonant/RWA limit

Define the detuning

$$\Delta_{F'F} \equiv \omega - \omega_{F'F}. \quad (31)$$

If $|\Delta_{F'F}| \ll \omega + \omega_{F'F}$ and the field remains perturbative, the counter-rotating term may be neglected. For a moving atom this gives

$$D_K \simeq \text{Re} \frac{1}{\omega_{F'F} - (\omega - kv_z) - i\gamma_{F'F}/2} = \text{Re} \frac{-1}{\Delta_{F'F} - kv_z + i\gamma_{F'F}/2}. \quad (32)$$

The same Doppler average then leads to Eq. (23). The counter-rotating term should be retained for a formally complete wide-band or static-limit treatment.

10. Practical calculation recipe

1. Choose isotope, nuclear spin I , ground $nS_{1/2}$ manifold F , and temperature T .
2. Tabulate every relevant hyperfine-resolved transition frequency $\omega_{n',J',F',F}$ and homogeneous linewidth.
3. Use exact hyperfine reduced dipoles in Eq. (8), or electronic reduced dipoles with Eq. (10).
4. For every F' separately, evaluate the Doppler-broadened response $\bar{D}_K$ of Eq. (20), equivalently the plasma-dispersion function in the rotating term.
5. Sum the $K = 0, 1, 2$ responses and convert to $\bar{\alpha}_F^{s,v,T}$ with Eq. (7).
6. Compute C_z and D_z from the polarization and evaluate Eq. (1), or diagonalize Eq. (17) if off-diagonal light couplings matter.
7. For a broadband source, integrate the frequency-dependent response against the measured spectral profile as in Eq. (29).

11. Checks and limiting cases

A correct implementation should satisfy:

- Reversing optical helicity sends $C_z \rightarrow -C_z$ and reverses only the vector shift.

- Linear polarization gives $C_z = 0$.
- The tensor shift vanishes at $|\epsilon_z|^2 = 1/3$ (the rank-2 “magic angle” condition).
- Under $m \rightarrow -m$, the scalar and tensor shifts are unchanged while the vector shift changes sign.
- If all excited hyperfine splittings are artificially set to zero for a ground $J = 1/2$ manifold, the rank-2 contribution cancels even after Doppler averaging.
- If the Doppler width greatly exceeds the excited hyperfine splittings, the tensor shift is strongly suppressed because the distinct F' dispersive profiles become nearly identical.
- In the far-off-resonant/static limit, retain the counter-rotating term.

12. Reference map

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Conceptual summary. The irreducible scalar contraction in Eq. (6a) is the rotationally invariant starting point for the fixed- F operator. Rewriting its $K = 0, 1, 2$ pieces in Cartesian form produces Eq. (17); the rank-1 piece becomes optical helicity coupled to spin orientation, while the rank-2 piece becomes light alignment coupled to the symmetric-traceless spin-alignment tensor. The rank- K decomposition is physical; the placement of numerical factors between $\alpha_F^{(K)}$, $\alpha_F^{s,v,T}$, the polarization tensors, and the normalized spin operators is conventional. In particular, the $1/2$ in $m/(2F)$ is not the induced-dipole $1/2$. Doppler broadening enters by thermally averaging each hyperfine-resolved optical denominator before the F' sum. Mathur–Tang–Happer’s δE_2 is precisely the rank-2/tensor light-shift operator in this language.